The Fiscal Incidence of Trade-Related Labour-Income Shocks in the UK*

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Abstract

I estimate how UK statutory tax–benefit cushioning differs between earnings-equivalent wage cuts and displacement under labour-income stresses associated with the 2025 US tariffs. I combine the 2024–25 Family Resources Survey, SIC exposure and PolicyEngine UK in a two-anchor, three-duration scenario design. OBR-style and unit 12-month-equivalent losses are £354 million and £886 million. At unit stress, cushioning is 43.9 per cent under wage cuts and 34.0 per cent under displacement; Exchequer costs are £484 million and £378 million. Across anchors and durations, wage-cut cushioning is 9.8–12.8 percentage points higher. Balanced and Bernoulli assignment estimates agree at the unit scale, although effective survey support remains thin. A public HMRC destination panel fails a tariff-intensity falsification, while calibrated longitudinal-LFS risk models agree weakly. The results are conditional statutory comparisons from a static partial-equilibrium model, not causal estimates of tariff transmission.

Keywords: tariffs, trade shocks, microsimulation, poverty, inequality, public finances

JEL codes: F13, F16, H23, I38, D31

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1 Introduction

This paper is a static, partial-equilibrium, first-round stress test of the UK tax–benefit system under alternative labour-income scenarios calibrated to sectoral tariff exposure. It is not a model or estimate of the economic transmission of tariffs. In a complete trade model, tariffs first change border prices and quantities, sourcing and market access; firms then adjust production, intermediate demand, value added, mark-ups, investment and productivity; only subsequently do labour demand, hours, wages, employment and household income respond. This paper does not estimate those links. It replaces them with an imposed reduced-form tariff-to-wage-bill bridge and alternative assumed adjustment margins, then evaluates the statutory tax–benefit consequences conditional on each scenario.

How differently does the UK tax–benefit system cushion a common earnings loss when it arrives through proportional wage cuts rather than job loss? That is the paper's primary question. The 2025 US tariffs provide the sector exposure and policy setting; they do not supply a natural experiment for the household comparison. A rich institutional literature traces the macroeconomic and firm-level consequences of the tariffs (OBR, 2025; Bank of England, 2025; City-REDI, 2025; Centre for Cities, 2025), while tariff studies map prices and income exposure into household welfare (Artuc et al., 2021; Luu et al., 2020). This paper instead isolates the statutory automatic-stabiliser response conditional on declared labour-income losses.

The calibration combines the tariff schedule, US export exposure by SIC division, and an export-demand scenario parameter to construct a sector exposure index. The exposure shares are built from ONS and HMRC data. A further wage-bill-incidence assumption maps that index directly to employee earnings. This deliberately strong step suppresses the intervening responses of export prices and quantities, domestic sales, imported inputs, production, productivity, profits and capital; it is not a structural production-side equation. Main evidence reports an OBR-style lower anchor of 0.4 alongside a unit stress of 1.0; the latter remains the common scale for the detailed margin decomposition. Sensitivities include the former April-outturn ratio of 2.0 as a high case and a severe value of 3.0. None is treated as an estimated coefficient. The resulting wage-bill stress is imposed on Family Resources Survey workers through four assumed adjustment families: wage cuts, displacement, inactivity among selected older workers, and reallocation into selected service industries at an FRS-calibrated earnings penalty. Because the pure wage-cut family implies a survivor-wage response far above empirical rent-sharing elasticities of 0.05–0.15 (Card et al., 2018; Garin and Silvério, 2024), it is read as an upper-bound stress scenario. A stylised mixed sensitivity puts 15 per cent of the imposed wage-bill loss on survivor wage cuts and 85 per cent on displacement; the numerical 15 per cent is literature-disciplined, but the split is not empirically identified by a rent-sharing elasticity. PolicyEngine UK then computes the conditional disposable-income, poverty, inequality and Exchequer responses. Employment and wages are scenario inputs at the microsimulation stage, not estimated consequences of the tariff.

The pre-specified submission design crosses OBR-style 0.4 and unit anchors, wage-cut and displacement margins, and three-, six- and twelve-month duration-equivalent annual stresses. At the unit twelve-month scale, wage cuts are cushioned by 43.9 per cent and displacement by 34.0 per cent, a 10.0-percentage-point difference. Exchequer costs are £484 million and £378 million. Across all six anchor-duration pairs, wage-cut cushioning is 9.8–12.8 points higher. Balanced and independent Bernoulli assignment estimates agree closely at the unit scale, but effective record support remains thin; assignment, risk-model and sector-exclusion sensitivities are reported separately from numerical Monte Carlo error.

Two sectoral nuances discipline the headline claims. Steel carries a caveat: roughly 80 per cent of UK steel exports go to the EU, so the steel–Wales cell is more exposed to EU/UK safeguards than to US tariffs. And pharmaceuticals—exempted under the December 2025 deal in exchange for VPAG rebate cuts and NHS pricing concessions—is a fiscal and pricing channel, not an employment channel, and is treated as such. Finally, an EPD counterfactual (full tariffs versus deal-mitigated rates) prices what the deal was worth, sector by sector and region by region, to households and the Treasury. Following Ignatenko et al. (2025), revenue recycling can flip aggregate employment signs; we compute first-round fiscal incidence holding fiscal policy fixed.

The remainder of the paper reviews the relevant evidence, defines the conditional scenario design, reports the primary contrast and concludes. Exploratory reallocation, outturn-shaped, supply-chain and geographical exercises are retained in the separate online supplement.

2 Literature

This paper sits at the intersection of four literatures: the quantitative and reduced-form work on the incidence of trade shocks; the evidence on how workers actually adjust to them; the institutional analyses of the 2025 US tariffs on the UK; and the microsimulation literature on tax–benefit cushioning of earnings shocks. Existing macro–micro studies impose labour-market shocks on tax–benefit models, while tariff studies map price and income exposure into heterogeneous household welfare. The World Bank's Household Impacts of Tariffs framework combines household consumption and income portfolios (Artuc et al., 2021); the OECD links its METRO trade model to household-budget surveys through an explicit GTAP–COICOP concordance (Luu et al., 2020). The narrower contribution here is a UK application that maps the 2025 partner-country export-demand exposure into alternative labour-income stress scenarios and then through the statutory tax–benefit system. It is a first-round scenario exercise, not a causal estimate of the tariffs' total household incidence.

2.1 Trade shocks and their incidence

Two recent papers anchor the structural end and frame the adjustment-margin question as its poles. Ignatenko et al. (2025) build a quantitative general-equilibrium model—nesting Armington, Eaton–Kortum, Krugman and Melitz structures with exact hat algebra over 194 countries—in which the 2025 tariffs enter as a *primitive* on trade shares. Wages are flexible and labour supply elastic, so employment is an equilibrium *outcome*; strikingly, its sign flips with revenue recycling (a tariff-financed income-tax cut raises US employment by 0.32 per cent, a lump-sum rebate lowers it by 0.41 per cent), and the optimal US tariff is a uniform 19 per cent independent of bilateral deficits. The UK is in their sample, and their replication data provide the UK-specific welfare and employment cells we use as aggregate anchors. Atkin et al. (2025), by contrast, apply a Baqaee–Farhi wedge framework to Chilean administrative data with factor supplies *fixed*: there is no employment margin at all, all adjustment runs through flexible wages and prices, and household incidence arrives via consumption baskets, factor ownership, profits and transfers—their "statutory versus economic incidence" framing is the one we adopt. One paper derives employment from the tariff primitive; the other assumes it fixed and lets wages absorb everything. A third structural anchor sits between the poles: Rodríguez-Clare et al. (2025) embed the 2025 trade war in a dynamic trade-and-reallocation model with *downward nominal wage rigidity*, in which higher tariffs raise US manufacturing employment but lower total employment as falling real wages depress participation, with US real income down about 1 per cent by 2028—rigid wages turn the same primitive into an employment shock that flexible-wage models turn into a wage shock, which is precisely the disagreement our adjustment-margin axis makes an explicit scenario dimension. On the partner-country side, Michelena et al. (2025) run the 2025 escalation through a multiregional input–output framework and find employment and export losses concentrated in the *targeted* countries—the same statutory-versus-economic distinction at the country level, but at the level of sectoral aggregates rather than households. Complementing the model-based work, den Besten and Känzig (2025) construct a narrative series of US tariff shocks over 1840–2024 and find tariff increases robustly contractionary—imports fall sharply, exports decline with a lag, and output drops persistently—historical discipline for treating the shock as a negative export-demand shock rather than a benign relative-price change.

The reduced-form lineage begins with Autor et al. (2013): import competition caused large, persistent employment and wage losses in exposed US commuting zones, with effects persisting to 2019 and no offsetting migration (Autor et al., 2021). Caliendo et al. (2019) provide the structural counterpart—a dynamic spatial model with costly mobility in which the China shock is a productivity/trade-cost primitive, generating aggregate gains alongside concentrated regional losses—and Adão et al. (2023) supply the theory-consistent mapping from regional shift-share estimates to aggregate counterfactuals that cautions against reading regional results as national ones. Kim and Vogel (2021) translate reduced-form employment and wage responses into money-metric worker welfare, finding regressive incidence with displaced workers' losses exceeding wage losses; Galle et al. (2023) show in a Roy-model setting that aggregate gains coexist with group-specific losses. Our contribution relative to this structural incidence work is to replace stylised worker groups and stylised transfers with real households and the actual UK tax–benefit system.

The 2018–19 US tariff episode supplies the sharpest direct evidence on who pays a tariff. Pass-through into US import prices was essentially complete, placing the incidence on the importing country's firms and consumers (Amiti et al., 2019; Fajgelbaum et al., 2020), with retail prices adjusting more slowly than border prices (Cavallo et al., 2021)—a pattern the 2025 wave repeats: matching daily retailer microdata to product-level tariff rates, Cavallo et al. (2025) estimate retail pass-through of about 20 per cent by September 2025, contributing roughly 0.7 percentage points to the US CPI; and Flaaen and Pierce (2019) show that US manufacturing employment fell on net, as input-cost increases and foreign retaliation outweighed import protection. The retaliation side of that episode is the closest analogue to our channel—a partner's tariff arriving as an export-demand shock: Waugh (2019) finds Chinese retaliatory tariffs depressed consumption in exposed US counties, and Blanchard et al. (2019) trace the electoral fallout of retaliatory tariffs across US counties. The broader distributional literature—surveyed in Fajgelbaum and Khandelwal (2016) and Fajgelbaum and Khandelwal (2022), with Borusyak and Jaravel (2021) decomposing expenditure against earnings channels—establishes that trade shocks redistribute along both consumption and income margins. The narrower contribution here is to apply a UK statutory tax–benefit microsimulation to alternative labour-income stress scenarios; it does not claim that the broader trade-and-household intersection is empty. Two propagation channels our direct-channel design deliberately excludes have well-measured analogues in this literature. Firm-level shocks travel along supply chains—Barrot and Sauvagnat (2016) show idiosyncratic disruptions imposing substantial output losses on customers of affected suppliers, and Acemoglu et al. (2016) find network effects on aggregate outcomes several times larger than the direct effect—and tradable-sector job losses can affect local services (Moretti, 2010; Gathmann et al., 2020). These estimates are not mechanically composable with our accounting sensitivity; Section 6 carries the caveat. Open-economy macro adds an offsetting margin: exchange rates absorb part of a tariff, but imperfectly and state-dependently—at most a fifth of the 2018–19 dollar appreciation is attributable to the tariffs (Jeanne and Son, 2024), the dollar appreciates on unilateral tariffs but depreciates under retaliation (Corsetti et al., 2025), and tariff *uncertainty* itself depreciated the dollar in 2025 through a risk-premium channel (Kalemli-Özcan et al., 2026); model-based analyses of retaliation similarly find non-trivial consumption losses for passive trade partners (Auray et al., 2020). We hold the exchange rate fixed; no scenario parameter is claimed to absorb or identify that response. Early assessments of the 2025 wave itself (Fajgelbaum and Khandelwal, 2026) again find the incidence landing on the tariff-imposing country's consumers—complementary to our question, which is the incidence on the tariffed *partner's* workers.

2.2 Adjustment margins: how workers absorb trade shocks

The worker-level evidence says the margin is mostly earnings, not permanent unemployment. Autor et al. (2014) find US workers exposed to import competition lose earnings amid churn across employers and sectors rather than through indefinite joblessness. Traiberman (2019) shows for Denmark that occupation-specific human capital produces reallocation with persistent wage losses; because our shock is an *export-demand* shock rather than import competition, the most direct microeconometric precedents are Hummels et al. (2014), who instrument Danish firm-level export demand and trace it into worker wages by task, and Garin and Silvério (2024), who show that Portuguese exporters pass idiosyncratic export-demand shocks into the wages of incumbent workers rather than shedding them—evidence for including a wage-cut dimension in our scenario grid. The rent-sharing literature disciplines the plausible scale of incumbent wage responses: Card et al. (2018) survey elasticities of wages to firm-specific rents of 0.05–0.15, and Kline et al. (2019) estimate that workers capture roughly 30 per cent of patent-induced surplus, with gains concentrated among incumbent men. Against that evidence, our pure wage-cut and pure displacement margins are polar stress scenarios. The 15/85 mixed point of Section 3.4 is literature-disciplined but is not an empirical conversion of those elasticities into adjustment shares. Dauth et al. (2021) provide the cleanest evidence for the wage-margin family, with German workers reallocating into services and other plants at lower wages. Dix-Carneiro (2014) estimates that reallocation takes years, with adjustment costs of 1.4–2.7 times annual wages and older workers hit hardest—calibrating our transition horizons and age gradients—while Dix-Carneiro and Kovak (2017) show regional effects that *grow* over two decades on a joint wage/employment margin.

The UK's own history points to a third margin absent from the US and German studies: benefit-financed economic inactivity. Beatty et al. (2007) document that coalfield job losses became male inactivity on incapacity benefits, not measured unemployment, with incomplete recovery after twenty years; Beatty and Fothergill (2017) show that 1980s–90s industrial job destruction *still* inflates the deficit through higher benefit claims and lower tax receipts—the direct UK precedent for our fiscal-incidence question, retrospective and area-level where ours is prospective, rules-based and household-level. The margin is not merely historical: Roberts and Taylor (2022) show in 2009–18 longitudinal data that disability benefit claims still respond to local labour-market slack over and above health status, Beatty and Fothergill (2023) document the persistence of hidden unemployment among incapacity claimants across large parts of Britain, and the IFS records 4.2 million working-age people on health-related benefits by 2024, up from 3.2 million in 2019 (IFS, 2024)—the inactivity margin is live at exactly the moment the tariff shock arrives. Aragón et al. (2018) add gender-differentiated margins from mine closures, and Rud et al. (2022) decompose displacement costs into wage cuts on re-employment versus non-employment spells—directly calibrating the split between our displacement and wage-cut families. For UK-specific elasticities, De Lyon and Pessoa (2021) is the key source: UK workers in Chinese-import-competing industries lost employment years and earnings without equally gainful re-employment, with larger female losses through employment years (see also Dorn and Levell, 2024). Closest to our object, Irastorza-Fadrique et al. (2025) study UK *household* responses to trade shocks—added-worker effects, later retirement, self-employment—but model behaviour, not tax–benefit mechanics; we treat their margins as complementary to our rules-based incidence.

2.3 The 2025 tariffs and the UK

The institutional literature on the 2025 shock is rich but stops above the household. NIESR's NiGEM scenarios and the OBR's March 2025 tariff analysis provide the macro anchors—the OBR puts the permanent UK GDP loss from a 20-point US tariff at one-third to three-quarters of a per cent (NIESR, 2024; OBR, 2025). The Bank of England estimates the effective US tariff on the UK at roughly 9 per cent (against about 18 per cent globally) and finds trade diversion has mildly *lowered* UK import prices (Bank of England, 2025; Dhingra, 2025)—which is what licenses our isolating the export-demand/earnings channel while holding the price and monetary channels fixed; the Bank's own staff work on the exchange-rate response to tariffs and retaliations (Corsetti et al., 2025) supplies the FX side of that held-fixed channel. The multilateral institutions frame the aggregates: the IMF's October 2025 outlook has global growth slowing to 3.2 per cent in 2025 amid the tariff shock (IMF, 2025), and the WTO's October 2025 update puts merchandise trade volume growth at 2.4 per cent in 2025 but only 0.5 per cent in 2026 as the tariffs bind (WTO, 2025). Firm surveys find most UK firms expecting small negative effects, with the US taking about 22 per cent of UK exports (CEP, 2025). The regional and sectoral geography is sharp: Coventry sends 22 per cent of its exports to the US (Centre for Cities, 2025); auto tariffs are estimated to cost £10,000 million of GDP over five years with the West Midlands bearing 62 per cent of it (City-REDI, 2025); SMMT data show US vehicle exports collapsing from 7,077 units in April to 4,223 in May 2025 before the EPD's implementation (SMMT, 2025). UK Steel notes that some 80 per cent of UK steel exports go to Europe—the basis of our steel caveat—and the December 2025 pharmaceutical deal (0 per cent tariffs for three years, paid for through VPAG rebate cuts from 22.5 to 14.5 per cent and NHS pricing concessions) makes pharma a fiscal/pricing shock rather than an employment shock (ABPI, 2025). The Resolution Foundation's household-facing analysis is descriptive rather than microsimulated (Resolution Foundation, 2025a); the IFS has no dedicated distributional publication on the 2025 tariffs. The closest work leaves room for a narrower UK statutory tax–benefit stress-test contribution.

For UK trade-shock precedents more broadly, Dhingra et al. (2017) map Brexit trade costs to income losses, and Fetzer and Wang (2024) provide the best district-level trade-shock incidence estimates (synthetic-control output losses of 5–10 points concentrated in manufacturing districts); Fetzer (2019) and Bloom et al. (2019) supply the political-economy and uncertainty channels. On household prices, Breinlich et al. (2022) use product-level import shares to identify the pass-through of the referendum depreciation, estimating a 2.9 per cent consumer-price increase and a similar real-wage loss. Levell et al. (2017) show why incidence can differ across UK households: food has much greater import content than consumption overall and household budget shares differ. These studies benchmark a consumption-price channel that the present cash-income exercise deliberately holds fixed.

The institutional modelling benchmark is also relevant to what this exercise does and does not claim. The UK Trade Modelling Review recommends a relatively simple, empirically grounded CGE core, complemented by granular partial-equilibrium modules, regional and dynamic extensions, parameter ranges and ex-post validation (Department for International Trade, 2021). In that architecture the tariff is a policy primitive: relative prices and trade quantities adjust first, production and sectoral value added respond next, and factor demand, wages and employment emerge jointly with reallocation across activities. Firm heterogeneity adds selection and productivity responses, while input–output structure transmits intermediate-cost and demand shocks. The OECD METRO model embodies this sequence in a multi-sector, multi-region production system before mapping induced prices into household expenditure and compensating variation (Luu et al., 2020). The World Bank instead offers transparent first-order household accounting of consumer and producer-income exposure, while explicitly excluding substitution, variety and productivity responses that can dominate beyond the short run (Artuc et al., 2021). Relative to those economy-wide or price-side tools, this paper supplies only the terminal tax–benefit module: it conditions on an imposed labour-income distribution rather than deriving it from production, productivity or labour demand.

2.4 Tax–benefit cushioning of earnings shocks

The methodological ancestors are Dolls et al. (2012), who use EUROMOD and TAXSIM to show EU tax–benefit systems absorb on average about 38 per cent of a proportional income shock and 47 per cent of an unemployment shock (against roughly 32 per cent in the US). The EU averages are not the right comparator for this paper; their *UK* cells are, and they are close to ours: the UK income-shock stabilisation coefficient is 0.352 (income tax 0.267, social insurance contributions 0.054, benefits 0.031) and the UK unemployment-shock coefficient 0.415 (tax 0.191, SIC 0.061, out-of-work benefits 0.163). Our 43.9 per cent wage-cut rate and 36.8 ± 4.9 per cent displacement rate are of a similar order, which is a useful benchmark rather than an external validation test. Section 4.2 instead focuses on the *composition* of the stabilisation total and the exposed-worker population. The interpretation of the tax term as an effective *marginal* rate is theirs too—and older still: Auerbach and Feenberg (2000) originate the framing of the income tax as an automatic stabiliser operating at marginal rates, and McKay and Reis (2016) formalise how progressivity and the extensive/intensive structure of shocks determine stabilisation in general equilibrium. The asymmetry between marginal and average relief is therefore not our discovery; our application shows how that asymmetry operates for the selected exposed-sector population. A second ancestor is Sutherland and Figari (2013) on EUROMOD's stress-test use—but both apply stylised macro shocks, not trade-policy-derived ones. Our direct methodological sibling is Brewer and Tasseva (2021), who pass the Covid-19 labour-market shock through UKMOD to show that the UK tax–benefit system plus the emergency support schemes held the rise in income inequality far below the rise in earnings inequality: the same architecture—an externally derived shock realised on survey households and run through the full UK benefit rules—applied to a pandemic where ours applies it to a trade-policy shock, without the emergency schemes and with the adjustment margin as the central axis. The wider macro-micro linkage literature supplies the lineage for the shock-derivation step itself: Bourguignon et al. (2008) codify the techniques for passing macro shocks to household microdata, Peichl (2016) surveys the linkage of microsimulation to structural macro models, and Navicke et al. (2013) show how EUROMOD nowcasts impose estimated labour-market transitions on survey microdata—the same device as our displacement draws, applied there to observed rather than counterfactual shocks. On the US side, exposed commuting zones saw rising disability, retirement and welfare payments that offset only a small fraction of earnings losses (Autor et al., 2013); Hyman et al. (2024) show wage insurance raises re-employment and earnings, a candidate counterfactual reform for this framework. The nearest tariff-microsimulation links are on the US side: the Tax Policy Center's tariff models allocate tariff burdens to tax units through consumption baskets in a full microsimulation model (Tax Policy Center, 2025), the Budget Lab at Yale prices the 2025 schedule's fiscal and distributional effects by income decile (Budget Lab, 2025), and Acosta and Cox (2024) document that the US tariff code is built regressively, with higher rates on lower-end varieties of the same goods. Two institutional facts motivate the cushioning asymmetry examined here. UC's capital rules—thresholds frozen in cash terms since 2006—reduced or extinguished entitlement for around two million income-eligible families, with roughly 1.1 million standing to gain from uprating alone (Resolution Foundation, 2025b); and take-up of income-related benefits is well below 100 per cent (DWP, 2024), so statutory replacement rates can overstate what displaced workers receive. Take-up gates entitlement inside the model and is explored only in a five-assignment sensitivity grid (Section 3.6), while the capital rules are a real-world gate the plain FRS data cannot exercise (Section 6). The distinction between consumer-price incidence and a partner's export-demand/earnings incidence defines the paper's narrower contribution: a UK statutory fiscal stress test organised around alternative adjustment margins. The results do not overturn the automatic-stabiliser literature; they show how the tax and benefit components differ across imposed margins for this exposed-sector population.

3 Methodology

3.1 Design: primitives first

Throughout, "derived" means generated by the stated scenario mapping, not causally estimated. The central demand parameter is a transparent unit stress, $\varepsilon = 1.0$, rather than an estimate. The grid includes $\varepsilon = 0.4$ as an OBR-style lower anchor, 2.0 as the former April-outturn high case, and 3.0 as a severe sensitivity. The April movement is not used to identify the centre because it also reflects anticipation, front-running, prices, exchange rates, composition and other demand conditions. Likewise, $\pi = 1.0$ is a deliberately strong full wage-bill-incidence assumption. The parameter grid is scenario sensitivity, not a statistical confidence interval.

A reader is entitled to ask, before any number is reported, which parts of this design are read off data and which are assumed. The answer, stated in full:

Object	Status
Tariff schedule τ_j	**Data** (policy documents: the April 2025 schedule, the May 2025 EPD, the December 2025 pharmaceutical agreement).
US-export intensity x_j	**Data** (HMRC RTS/OTS customs exports over ONS ABS division turnover; online supplement).
Export fall	**Data** in the measured family (realised HMRC outturn, May 2025–February 2026); **assumed** in the tariff families through declared low, central, high and severe demand scenarios.
Pass-through π	**Assumed** ($\pi = 1.0$), varied on a grid (online supplement).
Adjustment margin	**Assumed**, and deliberately varied. This is the paper's free dimension.
Household incidence	**Computed** from statutory tax and benefit rules in PolicyEngine UK; no behavioural or reduced-form step.

The margin is assumed because the literature does not agree on it, and the disagreement is structural rather than empirical: Ignatenko et al. (2025) derive the employment response from elastic labour supply and flexible wages, while Rodríguez-Clare et al. (2025) derive it from downward nominal wage rigidity, so the same tariff primitive arrives as a wage shock in the first model and a job shock in the second. We do not endogenise the margin. We price the disagreement—holding the aggregate earnings loss fixed and reporting what each pole implies for households and the Exchequer. That is a weaker claim than a structural model makes, and a more auditable one.

The shock enters through primitives, not through an assumed employment effect. The pipeline is:

$$s_j \; = \; \tau_j \, x_j \, \varepsilon \, \pi, \tag{1}$$

where s_j is an imposed employee-wage-bill stress for SIC 2007 division j, τ_j is its tariff rate, x_j is US-export intensity, ε is an export-demand calibration and π is a *wage-bill-incidence assumption*. Calling π pass-through would conflate distinct mechanisms. A structural transmission would separately map the tariff into export prices and quantities, quantities into domestic production and value added, and production into factor demand, productivity, profits, wages, hours and employment. Equation (1) compresses those links into one accounting bridge. It neither estimates a production function nor identifies the causal response of productivity, output or labour demand.

x_j is US goods exports as a share of division turnover, built from HMRC trade data and ONS Annual Business Survey turnover denominators; the online supplement documents the frozen provenance and per-division denominator choice. The central $\varepsilon = 1.0$ is a transparent unit stress, not an elasticity estimate; it amounts to a unit tariff-to-export-demand semi-elasticity, under which each percentage point of tariff removes one per cent of US-bound demand, and $\varepsilon = 0.4$ is the lower anchor, in the spirit of the trade-elasticity assumptions underlying the OBR's March 2025 tariff scenarios (OBR, 2025). The central $\pi = 1$ assigns the full calibrated exposure to the employee wage bill; it is intentionally severe and is not established by the economics literature. The sensitivity grid varies $\varepsilon\pi$, but this is parameter sensitivity, not an uncertainty interval around an identified coefficient. The implied gross earnings loss is therefore a conditional scenario input. OECD METRO and the UK Trade Modelling Review instead place tariff shocks inside multi-sector production systems in which prices, output, value added and factor demand adjust jointly (Luu et al., 2020; Department for International Trade, 2021); the World Bank HIT framework similarly labels its direct household calculations first-order and excludes productivity and substitution responses (Artuc et al., 2021). This paper begins only after those missing production-side mechanisms would have generated a labour-income distribution.

3.2 Public product–destination benchmark

To test whether post-policy US exports move differently from other UK export destinations, I build a monthly HMRC OTS panel from January 2018 to May 2026 for the United States, Canada, Japan and Australia (HM Revenue and Customs, 2026). Detailed HS8 rows are aggregated to HS4 and balanced across products, destinations and months, with absent flows set to zero. HMRC's separate two-digit aggregate rows are excluded. For product p and month t, the outcome is

$$g_{pt} = \log(1 + V_{p,\mathrm{US},t}) - \frac{1}{3} \sum_{d \in \{\mathrm{CA},\mathrm{JP},\mathrm{AU}\}} \log(1 + V_{pdt}).$$

I regress this destination gap on a post-May-2025 indicator, HS4 product fixed effects, month-of-year effects and a linear trend. April 2025 is omitted as an announcement and anticipation month. The primary WLS weights are products' mean 2022–24 US export values, capped at the 99th percentile; standard errors are clustered by HS4. Uncapped value weights, equal product weights, individual destinations, later sample starts and pseudo-policy dates in May 2019–23 are reported as specification checks.

This design controls for product shocks common across destinations but not US-specific demand, exchange rates, shipping, inventories, composition or other coincident policies. The high-tariff comparison groups HS chapters 72, 73 and 87 as steel and automotive products; that broad mapping is a falsification check, not a complete product-level statutory tariff schedule. The panel is therefore not used to estimate ε or π in Equation (1). It tests whether the imposed bridge is consistent with destination-differential outturn patterns.

3.3 The tariff schedule and the EPD counterfactual

Two tariff scenarios bracket the policy. *Full tariffs*: the April 2025 schedule—10 per cent baseline on UK goods, 25 per cent on motor vehicles (SIC 29) and steel (SIC 24). *EPD*: the deal-mitigated schedule—autos at the 10 per cent in-quota rate on up to 100,000 vehicles; steel with conditional, partial relief; pharmaceuticals (SIC 21) exempt under the December 2025 agreement. The automotive scenario applies the 10 per cent rate to the modelled 2024 exposure because the quota is approximately equal to that year's US-bound volume; it does not model quota allocation, above-quota growth or an endogenous volume response around the ceiling. The difference between the two scenarios, run through identical downstream machinery, prices the EPD for households and the Exchequer. Two caveats travel with the schedule. Steel: about 80 per cent of UK steel exports go to the EU, so the US tariff is a minority driver of the sector's exposure and the steel cell should be read alongside EU/UK safeguards. Pharmaceuticals: the exemption was purchased through VPAG rebate cuts and NHS pricing concessions, so pharma's incidence is fiscal and price-side, not an employment shock; we model its employment shock as zero under the EPD and discuss the fiscal channel separately (Section 5).

3.4 Adjustment-margin families

The derived s_j is realised on Family Resources Survey 2024–25 employees in exposed divisions. The submission design pre-specifies one primary contrast: statutory cushioning under displacement versus proportional wage cuts, conditional on a common expected sector wage-bill loss. Inactivity, reallocation, mixed margins and the EPD schedule are secondary sensitivities. Wage cuts remove $\sum_j s_j B_j$ deterministically; displacement removes that amount in expectation under the risk model and is numerically integrated using the balanced design below.

Displacement (ADH short run). Within each exposed division every employee has baseline risk s_j, irrespective of the record's survey weight. The primary estimator generates repeated random-start systematic candidates, ordered across age and within-industry earnings ranks, and retains the candidate that best matches the expected industry wage-bill and weighted- headcount losses while secondarily balancing broad age and earnings margins. This is balanced numerical integration, not a survey estimator: conditioning on aggregate balance intentionally changes exact record-level marginal inclusion probabilities. Selected workers move to unemployment with earnings, hours, pension contributions and statutory pay zeroed. An independent Bernoulli estimator, which preserves the declared first-order probabilities exactly but has much larger allocation dispersion, is reported as the unbiased assignment-rule comparator.

Wage cuts with reallocation (Dauth/Traiberman long run). No job loss; every employee in division j takes a proportional cut of s_j, so the weighted aggregate earnings removed equals $\sum_j s_j B_j$ by construction—exact earnings-equivalence with the displacement family's expected removal. Employee pension and salary-sacrifice contributions are scaled proportionally with the cut, mirroring their zeroing under displacement; this lowers taxable pay (and hence tax relief) but also reduces the household's pension outflow, so under the HBAI disposable-income concept part of the earnings loss is absorbed by reduced retirement saving rather than by current income.

Inactivity (Beatty–Fothergill UK history). The displacement draw, but displaced workers aged 50 and over exit to economic inactivity, the benefit-financed route of UK deindustrialisation, rather than unemployment. The route is benefit-financed in the model, not just relabelled: every inactive worker is flagged as having limited capability for work-related activity, so their household's Universal Credit includes the LCWRA health element (£217.26 a month in 2026—the reduced rate applying to new claims from April 2026 under the Universal Credit Act 2025; existing claimants retain the higher pre-reform rate, and all modelled inactivity claims are new claims). In policyengine-uk 2.89.2 the survey-imputed `uc_LCWRA_element` is stored, so the post-shock benefit-unit value is set to the maximum of the stored amount and one annual statutory element; an existing recipient is never paid a second element, unaffected units are unchanged, and the run hard-errors if the element fails to reach a newly flagged person's benefit unit. This mirrors the Beatty et al. (2007) route by which industrial job loss became incapacity-benefit inactivity. The strong assumption should be stated plainly: *all* displaced workers aged 50 and over are assumed to pass the Work Capability Assessment, so the family is an upper bound on the health-element route; no partial-qualification sensitivity (for example, 50 per cent of the older displaced qualifying) has been simulated, which is a limitation of the present scenario set. Work-search conditionality carries no separate fiscal machinery in the model, so the LCWRA element is the entire modelled wedge between inactivity and unemployment.

Reallocation into services (ADH/Autor et al., 2014). The same displacement draw on the same seed—so the two families are paired worker-for-worker—but the drawn workers are re-employed rather than left out of work. Their `sic_industry_division` is reassigned to one of four services destinations (SIC 47 retail, 86 human health, 49 land transport, 56 food and beverage service) in proportion to those divisions' FRS employee-weight shares (29.5, 45.8, 9.2 and 15.5 per cent), and their annual earnings are cut by a penalty calibrated on the FRS itself: weighted mean annual employee earnings of £48,272 over 2,047 hours in the exposed goods divisions against £34,594 over 1,701 hours in the four destinations, a **28.3 per cent annual-earnings penalty** that embeds the hours fall. Two variants are reported: an hours- and age-controlled **low-penalty** case of 14.0 per cent (the pure hourly-wage gap), and a **three-month lag** in which workers are out of work for a quarter of the year before the services job starts, scaling annual earnings and hours by a further $1 - 3/12$. ASHE 2024 full-time medians bracket the hours-controlled figure at 10–20 per cent; the FRS all-employee penalty is larger because it also captures the shift into part-time work. Because this family removes only the penalty on the movers rather than the full earnings-equivalent aggregate, its pound figures are comparable with the displacement family (identical worker sets, paired draws) but *not* with the wage-cut family; its cushioning *rate* is comparable with both. The online supplement reports the results and mechanics.

Mixed adjustment and factorial scenario testing. Let $\lambda \in [0, 1]$ denote the share of the sector shock delivered through displacement. A worker with sector shock s is displaced with probability $p = \lambda s$. Conditional on surviving, the proportional wage cut is

$$c = \frac{s - p}{1 - p},$$

so expected earnings loss is exactly $p + (1 - p)c = s$ for every worker. The endpoints reproduce pure wage cuts ($\lambda = 0$) and pure displacement ($\lambda = 1$). The exploratory grid crosses $\lambda \in \{0, .25, .5, .75, 1\}$ with export-demand calibrations $\varepsilon \in \{0.4, 1, 2, 3\}$ on five common assignments per cell. This follows the companion UK AI study's factorial principle: vary shock magnitude separately from adjustment composition. It is scenario testing, not a probability distribution over future outcomes.

The wage-cut margin as an upper bound, and a literature-disciplined mixed sensitivity. The pure wage-cut family should be read against the rent-sharing evidence, because it embeds a survivor-wage response far above anything estimated. Delivering the entire sector wage-bill loss through proportional cuts to incumbent workers' wages imposes a unit wage response to the sector exposure index. Garin and Silvério (2024) estimate an incumbent-wage elasticity of roughly 0.15 to firm-specific output shocks in Portuguese matched data; the rent-sharing survey of Card et al. (2018) places typical elasticities at 0.05–0.15; and Hummels et al. (2014) find small within-job wage responses in Danish data alongside losses through displacement. The wage-cut family ($\lambda = 0$) is therefore an *upper-bound stress scenario for survivor-wage incidence*, retained because it prices the flexible-wage pole of the structural disagreement.

The scenario set also reports a transparent 15/85 mixed sensitivity: 15 per cent of the imposed wage-bill loss is delivered through survivor wage cuts and 85 per cent through displacement ($\lambda = 0.85$). The value 0.15 is disciplined by the upper end of the cited wage-response estimates, but it is not an empirical calibration of the loss decomposition. A wage elasticity to firm output or rents is not algebraically the share of an aggregate wage-bill loss borne by survivor wages. Identifying that share requires linked employer–worker estimates of wages, hours, employment and re-employment. The mixed result is therefore a literature-disciplined scenario point, not the uniquely "most plausible" margin.

Two scope limits on the family set should be stated here rather than in a footnote. The wage-cut family models the *earnings consequence* of reallocation without moving anyone between sectors: workers keep their observed industry and hours, and the proportional cut stands in for the wage penalty that a completed reallocation would carry; the reallocation family does move them, but who reallocates is assumed (it is the displacement draw) rather than estimated. And no household-side behavioural margin is active—added-worker responses, earlier or later retirement, and shifts into self-employment (Irastorza-Fadrique et al., 2025) are all suppressed, as are any labour-supply responses to the tax and benefit changes themselves. All four families are therefore first-round, rules-based incidence.

A third caveat concerns selection within divisions. Export-exposed workers are unlikely to be representative of their division: exporters pay a wage premium and pass export-demand shocks into incumbent wages (Hummels et al., 2014; Garin and Silvério, 2024). Cushioning is nonlinear in earnings, so uniform risk may misstate its tax/benefit composition. The longitudinal-LFS cell, earnings-band and QRF sensitivities vary observed risk shape while recalibrating each division to its wage-bill target. They quantify selection- model sensitivity but cannot identify exporter status or tariff-specific worker risk.

The FRS support for the transition is thin. Of 34,966 person records, 12,800 have employee earnings and 1,018 are in mapped exposed divisions, but the reference probabilities imply only 9.7 selected records. The largest exposed record has a person weight above 19 thousand. Balanced integration constrains the realised aggregate and observed composition; it does not create survey information or establish population representativeness. I therefore report the affected record count, Kish-style effective number of gross-loss contributions, largest-record loss share, and balanced-versus-Bernoulli mean and SD comparisons. Subgroup rankings remain exploratory.

Duration-equivalent annual stresses. The annual PolicyEngine model cannot represent a worker employed for part of a year and unemployed for the remainder with internally consistent monthly UC assessment, work status and earnings. I therefore do not simulate three- or six-month unemployment spells. Instead I scale s_j by $d/12$, for $d \in \{3, 6, 12\}$, and assign a correspondingly smaller number of coherent full-year displaced states. These rows preserve the expected worker-month earnings loss associated with each duration while changing the extensive-margin stock. They are labelled *duration-equivalent annual stresses*, not spell estimates. Wage-cut rows use the same scale, preserving the common- loss comparison at every duration and at both the OBR-style 0.4 and unit anchors.

The margin is the paper's central axis because the tax–benefit system treats the two poles asymmetrically: an in-work earnings reduction is offset automatically at marginal tax and taper rates, while a job loss moves the worker to the gated, means-tested extensive margin of the benefit system. How much each pole is cushioned, and—the question that turns out to discriminate between them—how much that cushioning depends on anyone doing anything, are empirical questions the microdata answer (Section 4.2).

3.5 Longitudinal LFS imputation benchmark

The uniform within-division assignment in the central displacement family is transparent but suppresses observable worker heterogeneity. I therefore use UK Data Service Study 9490, the Labour Force Survey Five-Quarter Longitudinal Dataset covering July 2024–September 2025, as a donor benchmark. The donor population comprises respondents employed at wave 1. Job exit is observed at wave 5; log wage change is defined only for respondents employed with positive main-job pay at both endpoints. Negative UKDS sentinel values are missing, not zero. Public 2024 Nomis BRES employee totals rescale the manufacturing SIC composition before direct transition targets are computed. (Office for National Statistics, 2025)

The primary imputation maps SIC-division–sex–age cells to employed FRS receivers, shrinking sparse cells towards sector and manufacturing means. After matching, predicted exit risks are rescaled so their FRS-weighted mean equals the directly measured, BRES-composition-adjusted LFS manufacturing exit rate. Mean predicted log wage changes are centred on the corresponding LFS target. An income-tercile estimator supplies a transparent sensitivity specification.

A regularised quantile/random-forest benchmark follows the calibrated imputation architecture in PolicyEngine's `young-worker-nics` project. It estimates model shape on all wave-1 employed adult donors using age, sex and annualised employment income. The binary outcome uses the random-forest event probability; the continuous outcome uses the QRF conditional mean. Forests use 500 trees and a minimum leaf size of 20; wage outcomes are winsorised at the donor 1st and 99th percentiles. Both receiver predictions are then calibrated to exactly the same manufacturing LFS targets as the cell estimator. This is a predictive baseline-transition benchmark, not an estimate of tariff-induced exits or wage changes. It is therefore reported as a robustness diagnostic and does not replace the declared central assignment rule.

In a downstream allocation sensitivity, each receiver score r_i is converted to a within-division displacement probability $p_i = \min(1, k_j r_i)$, where k_j is solved numerically so

$$\frac{\sum_{i \in j} p_i E_i w_i}{\sum_{i \in j} E_i w_i} = s_j.$$

Thus cells, earnings bands and QRF change who is exposed while preserving each division's expected employee-wage-bill loss. Missing scores use the division mean. Common assignments then isolate sensitivity to incidence shape; they do not give the baseline transition model a causal tariff interpretation.

3.6 Universal Credit take-up after the shock

The take-up experiment is applied symmetrically across margins. After each shock, changed benefit units whose all-claim UC award moves from zero at baseline to positive post-shock receive a claiming draw at the scenario take-up rate. This includes wage-cut households that cross into entitlement. Existing claimants, existing eligible non-claimants, unchanged units and post-shock-ineligible units retain their baseline flag, avoiding an unintended population-wide take-up reform. The grid therefore measures sensitivity to a common new-entitlement convention; it does not estimate actual claiming behaviour.

One modelling choice deserves its own subsection, because an earlier version of this paper got it wrong and the correction changes a headline number. PolicyEngine UK carries benefit take-up as a benefit-unit-level *stored input*, `would_claim_uc`, drawn once at dataset build time. Two properties of that draw matter. First, its target is a flat population-wide parameter—0.55 as a share of *all* benefit units, with reported UC recipients anchored to TRUE and the remainder filled to hit the target. It is not a take-up rate among the entitled, and it is therefore not comparable to the roughly 80 per cent take-up-among-entitled that the Resolution Foundation and the IFS assume; in the dataset used here the realised weighted flag rate is 0.547 over all benefit units and 0.541 among UC-eligible ones. Second, the flag is drawn on *pre-shock* circumstances. Nothing in a naive shock pipeline re-draws it, so a worker displaced by the shock carries into unemployment the claiming decision attributed to them while they were employed and, in the great majority of cases, not entitled to anything—measured take-up among the displaced post-shock was 46.9 per cent, below even the population flag rate. The baseline flag is thus not informative about the post-shock claiming behaviour of a newly entitled family: it encodes a decision taken in a state of the world the shock has destroyed.

The pipeline therefore re-draws `would_claim_uc` after each shock only for benefit units whose circumstances changed and whose all-claim UC award moves from zero at baseline to positive post-shock, at a scenario parameter `uc_takeup` with default 0.80. The re-draw uses an independent random stream, leaving worker assignment unchanged and preserving paired comparisons. This rule applies to displacement, inactivity, reallocation and wage cuts alike; existing eligible, post-shock-ineligible and unchanged units retain their baseline flag. Take-up sensitivity is therefore evaluated symmetrically for newly entitled units, although the parameter remains assumed rather than observed.

3.7 Microsimulation and outcomes

The Monte Carlo draws describe sensitivity to the artificial assignment of displacement across weighted FRS records. Their standard deviations are assignment dispersion, not sampling standard errors, confidence intervals, or uncertainty about the tariff response, survey design or model parameters. Numerical Monte Carlo error of a reported mean is the assignment SD divided by the square root of the number of draws. Parameter scenarios vary the demand calibration, wage-bill incidence, take-up and adjustment composition, but are not assigned probabilities and do not form confidence intervals. The FRS complex-survey sampling variance and uncertainty in the trade-to-earnings bridge are not estimated. Reported comparisons are therefore descriptive scenario contrasts. Distributional cash-income changes use annual household disposable-income change divided by household size before person-weighted aggregation; aggregate disposable-income totals remain household-weighted totals.

The realisation step—an imposed status and earnings transition on survey households, then run through the full benefit rules—follows Brewer and Tasseva (2021), who apply it to the Covid-19 labour-market shock in UKMOD, and the EUROMOD nowcasting tradition of imposed labour-market transitions (Navicke et al., 2013). PolicyEngine UK computes baseline and shocked household disposable income (HBAI cash concept throughout), relative before-housing-costs poverty measured against a poverty line frozen at 60 per cent of the baseline equivalised median (so the reported change reflects households crossing a fixed threshold rather than mechanical movement of the line itself), absolute before-housing-costs poverty against a fixed baseline line, after-housing-costs poverty, the Gini coefficient, distributional breakdowns and the Exchequer effect. The *Exchequer effect* is defined as the change in the aggregate simulated government balance—all modelled tax and contribution revenue (including employer National Insurance contributions) minus all modelled benefit spending—between the baseline and shocked simulations. It is therefore broader than the employee-earnings loss ΔE that enters the cushioning rate; the online supplement reconciles the two. On timing: trade exposure is built from 2024 HMRC customs data, the household input is FRS 2024–25, and the simulation year is 2026, with PolicyEngine UK's standard uprating applied to survey monetary variables between the survey year and the simulation year. Primary submission scenarios use 50 balanced assignments; legacy direct and measured scenarios use 100 assignments, and explicitly exploratory diagnostics use five. Following Ignatenko et al. (2025), revenue recycling can flip aggregate employment responses; here fiscal policy is held fixed. Only direct taxes and benefits respond. Indirect taxes are not shocked, so reduced consumption-tax receipts are absent from the government balance.

4 Results

All results use FRS 2024–25 and period 2026. The primary table uses 50 balanced assignments; legacy secondary scenarios use 100. Values after ± are assignment standard deviations, not standard errors or confidence intervals. Numerical Monte Carlo SE is stored separately in each JSON artifact. Five-assignment diagnostics are exploratory.

4.1 Derived sector shocks

The deterministic OBR-style lower-anchor stress is £354 million annually. The unit-stress full-schedule amount is £886 million and the EPD unit stress is £693 million. These are scenario inputs generated by Equation (1), not estimated causal effects. The unit-stress sector rates are exactly half those of the former $\varepsilon = 2$ high case and 2.5 times the OBR-style rates.

4.2 Cushioning

Table 1: Primary common-loss contrast by anchor and duration-equivalent annual stress, 50 assignments

Anchor	Months	Margin	Gross loss (£m)	Exchequer (£m)	Cushioning (%)	Poverty (pp)
OBR	3	Displacement	89 ± 2	35 ± 8	30.9 ± 9.0	0.003
OBR	3	Wage cut	89	48	43.8	0.000
OBR	6	Displacement	177 ± 3	73 ± 13	32.7 ± 7.7	0.009
OBR	6	Wage cut	177	96	43.7	0.000
OBR	12	Displacement	354 ± 4	141 ± 21	32.0 ± 5.7	0.015
OBR	12	Wage cut	354	192	43.6	0.000
UNIT	3	Displacement	222 ± 4	91 ± 15	32.0 ± 6.2	0.008
UNIT	3	Wage cut	221	120	43.6	0.000
UNIT	6	Displacement	442 ± 6	190 ± 24	33.8 ± 5.4	0.016
UNIT	6	Wage cut	443	241	43.6	0.000
UNIT	12	Displacement	885 ± 6	378 ± 37	34.0 ± 4.7	0.036
UNIT	12	Wage cut	886	484	43.9	0.000

Notes: OBR uses export-demand elasticity 0.4; unit uses 1.0. Values after ± are assignment SDs. Displacement uses balanced numerical integration; wage cuts are deterministic apart from the common UC claiming redraw. Three- and six-month rows scale the annual probability of a coherent full-year displaced state; they match expected worker-month loss but do not simulate partial-year unemployment spells. Poverty is the change in the person-weighted fixed-line BHC rate.

Table 1 is the pre-specified primary evidence: it crosses the external OBR-style and deliberately stronger unit anchors with one common- loss contrast and three duration-equivalent scales. At the 12-month unit stress, wage-cut cushioning exceeds displacement cushioning by 10.0 percentage points (assignment SD 4.7), while its Exchequer cost differs by £107 million. These are conditional statutory comparisons, not confidence intervals for tariff effects.

Assignment-design and record-support audit. At the unit 12-month stress, balanced and Bernoulli integration agree closely: cushioning is 34.0 versus 33.5 per cent and Exchequer cost is £378 versus £377 million. Balancing reduces gross-loss SD to 0.02 of the Bernoulli value and Exchequer SD to 0.18, but it cannot repair sparse survey support. The unit displacement loss has only 6.2 effective record contributions and its largest record supplies 28.7 per cent on average. At the OBR three-month scale these deteriorate to 1.2 and 91.9 per cent. The smallest displacement rows are consequently sensitivity evidence, not precise population estimates. Across leave-each-sector-out reruns for all 23 exposed divisions, the wage-minus-displacement cushioning difference remains positive, ranging from 7.6 to 11.3 percentage points. No single division creates or reverses the primary contrast.

Part of the wage-cut margin's higher measured cushioning reflects a pension-contribution channel rather than the tax–benefit system alone: employee pension and salary-sacrifice contributions are scaled proportionally with the earnings cut, so part of the gross earnings loss is absorbed by reduced retirement saving rather than by current disposable income, while the associated tax relief falls. This is a measurement choice—the HBAI disposable-income concept treats employee pension contributions as an outflow—and it carries a caveat: the implied loss of deferred consumption in retirement is not captured by the cushioning rate.

Mixed-margin sensitivity. The stylised mixed scenario of Section 3 (survivor-wage share 0.15, displacement share 0.85) yields a cushioning rate of 38.3 ± 4.2 per cent and an Exchequer cost of £428 ± 221 million under the full schedule, and 38.0 ± 4.7 per cent and £336 ± 198 million under the EPD schedule. Both sit between the polar margins but much closer to displacement, mechanically reflecting the imposed 85/15 split. The result is a literature-disciplined sensitivity point, not an estimated pass-through or adjustment decomposition.

4.3 External magnitude and outturn checks

The frozen HMRC customs build totals £55.6 billion of 2024 US-bound goods exports before exclusions and maps £52.5 billion to producing divisions. This is reasonably close to the ONS balance-of-payments headline of £59.3 billion, given known conceptual and coverage differences (ONS, 2025). The reference full-schedule calibration predicts a 12.57 per cent aggregate fall across mapped divisions. The observed May 2025–February 2026 customs window falls 12.71 per cent against its two-year same-month baseline. This near equality is not a validation result: the reference calibration was not estimated from that window, and the outturn combines tariff effects with prices, exchange rates, anticipation, composition, rerouting and other demand shocks.

Sector agreement is much weaker. The stored validation artifact reports a rank correlation of 0.00 between the full-schedule predicted and realised falls among the ten largest exposed divisions, and a correlation of 0.58 between full-schedule and observed downside shock rates across all mapped divisions. Machinery and motor vehicles decline broadly in the expected direction, while aerospace, electrical equipment and basic metals do not. The aggregate match can therefore be coincidental and must not be used to validate Equation (1). As a separate policy benchmark, the OBR's March 2025 scenario uses a demand elasticity of 0.4, implying an 8 per cent export-demand fall after a 20 percentage-point tariff and a direct export-demand effect just under 0.2 per cent of GDP (OBR, 2025). The paper's $\varepsilon = 1$ reference is 2.5 times that elasticity and additionally assumes full wage-bill incidence; it is a deliberately strong stress, not a central forecast.

Product–destination panel. Across 1194 HS4 products, the capped value-weighted specification estimates a -31.9 per cent change in exports to the US relative to the same products sent to the three comparison destinations (95 per cent interval -44.8 to -16.0 per cent, $p =< 0.001$). Results have similar signs against each destination, and May 2019–23 pseudo-policy estimates are individually insignificant. This is not robust evidence of a common product effect: equal product weights give -4.9 per cent ($p = 0.293$), and restricting the sample to 2023 onward gives -17.5 per cent ($p = 0.080$). The estimated differential trend is itself borderline ($p = 0.057$). Most importantly, broad steel and auto chapters fall -20.0 per cent ($p = 0.290$), compared with -32.9 per cent for other products ($p = 0.001$): the tariff-intensity ordering fails. The online supplement displays the monthly gap. The panel supports a post-policy decline concentrated in large US export products, but it cannot attribute that decline to tariffs or validate the tariff-rate mapping.

Longitudinal worker benchmark. The five-quarter LFS contains 177 wave-1 manufacturing employees after the required population filters, compared with 1102 employed manufacturing FRS receivers. Its BRES-composition-adjusted manufacturing targets are an 8.96 per cent wave-1-to-wave-5 job-exit rate and 5.87 per cent mean log wage growth among valid endpoint-pay observations. Table 2 compares receiver distributions. Calibration deliberately makes the aggregate exit means identical, so the comparison tests incidence shape rather than level. The QRF has a much wider risk distribution and agrees weakly with the primary cells: person-level correlations are only 0.06 for exit risk and 0.08 for predicted wage change. The longitudinal evidence therefore does not support replacing transparent cells with forest-generated tariff incidence. It instead quantifies model uncertainty that the central uniform assignment omits. Regularisation does not eliminate all forest tails: the maximum calibrated exit risk is 62.6 per cent and the predicted log-wage-change range is -45.7 to 72.0 per cent. These are benchmark predictions, not plausible confidence bounds.

Table 2: Calibrated FRS job-exit-risk distributions

Estimator	Weighted mean	P10	Median	P90
Calibrated SIC–sex–age cells	8.96	7.15	8.42	11.88
Calibrated income terciles	8.96	3.88	8.91	13.77
Regularised QRF benchmark	8.96	1.22	5.42	19.62

Notes: Entries are percentages among matched employed manufacturing FRS receivers. All means equal the same directly measured LFS target by construction. Quantiles are survey-weighted. QRF denotes the regularised quantile/random-forest benchmark described in Section 3.5; it predicts baseline transitions, not causal tariff effects.

Fiscal sensitivity to worker-risk shape. Table 3 uses each calibrated risk model to vary selection within SIC divisions while holding every division's expected weighted wage-bill loss at the original shock. The cell model nearly reproduces uniform assignment. Across all three

LFS-shaped models, mean cushioning ranges from 36.0 to 37.4 per cent, Exchequer cost from £384 million to £425 million, and measured poverty change from 0.024 to 0.028 percentage points. This model spread is smaller than the assignment SD within any specification. Weighted displaced headcount changes because calibration preserves the wage bill, not a headcount quota. The QRF mean gross loss is £847 million rather than the £886 million expectation, but its £397 million assignment SD implies a numerical MC SE of about £40 million, so this difference is not evidence of a different aggregate target.

Table 3: Full-schedule fiscal sensitivity to within-SIC worker selection

Selection model	Workers (000s)	Gross loss (£m)	Exchequer (£m)	Cushioning (%)	Poverty (pp)
Uniform central assignment	17.2 ± 6.9	882 ± 418	412 ± 225	36.8 ± 4.9	0.028 ± 0.019
LFS calibrated cells	17.2 ± 6.9	881 ± 412	412 ± 223	36.9 ± 5.0	0.028 ± 0.019
LFS income terciles	15.2 ± 6.5	887 ± 467	425 ± 258	37.4 ± 6.0	0.025 ± 0.017
LFS regularised QRF	18.1 ± 5.9	847 ± 397	384 ± 212	36.0 ± 5.7	0.024 ± 0.016

Notes: Means ± assignment SD over 100 common assignments. Within each SIC division, individual probabilities are rescaled so the earnings-weighted mean equals the declared sector shock. Poverty is relative BHC against the frozen baseline line. LFS risks are predictive baseline-transition scores, not tariff-specific causal probabilities.

The realised displacement output is also consistent with the allocation design but too sparse for precise empirical incidence: the expected 9.7 selected FRS records per assignment produce 17.2 thousand represented displaced workers and £882 ± 418 million of gross loss. The gross-loss SD is nearly half its mean, and Exchequer-cost SD exceeds half its mean. The 100-assignment average is numerically more stable than an individual assignment, but it remains conditional on the observed 1,018 exposed records and the uniform within-division selection rule.

Take-up sensitivity. The regenerated grid applies the same post-shock-entitlement claiming rule to all margins. It uses only five assignments per cell and is exploratory. It shows that displacement responds more visibly to the assumed claiming probability, while wage-cut results move little because relatively few wage-cut households change UC entitlement. No grid difference is treated as statistically identified.

4.4 Mechanism

The regenerated decomposition and gate audit are also five-assignment diagnostics. Income-tax relief is larger on the wage-cut margin than on displacement; UC is more important after displacement. The plain FRS input contains no usable household-capital measure, so the model cannot establish the empirical importance of UC's capital test. New-style JSA is not activated by the imposed transition. These omissions make the gate audit a description of the implemented model, not the full UK out-of-work system.

4.5 Distributional incidence

Under full displacement, relative BHC poverty rises by 0.028 ± 0.019 percentage points. Under wage cuts it is unchanged at displayed precision. These national changes are small relative to omitted uncertainty and are not headline estimates. They can coexist with large selected-household losses because exposure is limited and concentrated above the poverty line. Decile profiles and assignment dispersion are reported in the online supplement.

4.6 Fiscal incidence

Annual Exchequer costs are reported in Table 1. The measure is the aggregate simulated government-balance change and is broader than the household cushioning identity. Employer National Insurance contributions and the tax consequences of pension and salary-sacrifice changes respond to the shock but sit outside employee disposable income. The online supplement provides the formal reconciliation and draw distributions.

4.7 EPD counterfactual

On paired seeds, full minus EPD displacement differs by 3.7 ± 2.8 thousand workers and £96 ± 101 million of Exchequer cost. These are secondary scenario contrasts, not causal deal estimates; detailed gross-loss, poverty and graphical results are in the online supplement.

5 Policy

5.1 Economic Prosperity Deal

Within the maintained direct-channel scenarios, the paired full-minus-EPD displacement contrast is 3.7 ± 2.8 thousand workers, £204 ± 192 million of gross earnings and £96 ± 101 million of annual Exchequer cost. These SDs describe assignment dispersion and the contrasts are not causal estimates of the deal. Autos drive much of the difference; the scenario applies the in-quota rate to fixed exposure without modelling quota allocation or future above-quota trade.

5.2 Pharmaceutical channel

The EPD scenario sets the pharmaceutical employment shock to zero, but this does not make the exemption free. Its VPAG and NHS pricing concessions sit outside the labour-income microsimulation. A complete policy appraisal would cost those health-budget and price effects alongside the direct tax–benefit scenario.

5.3 Cushioning policy

The simulations support a limited conclusion: extensive-margin protection depends more on means-tested eligibility and claiming than the income-tax response to an intensive earnings loss. The five-assignment take-up grid is exploratory and does not quantify claimant behaviour precisely. Automatic enrolment after redundancy, contribution-based unemployment insurance and wage insurance are therefore candidate counterfactuals for future analysis, not policy recommendations established by the present results. Their costs and incidence require dedicated simulations and financing assumptions.

The repository also supplies reusable interfaces for the next policy exercises. Observable household and worker characteristics can tilt shock probabilities while preserving each sector's aggregate earnings loss, and a separate price-channel interface converts expenditure-weighted price changes into real disposable income. Neither interface supplies causal coefficients: the former requires externally estimated heterogeneity and the latter requires household expenditure data. They are extensions to populate and validate, not additional headline estimates in this paper.

6 Discussion and conclusion

This paper pre-specifies a conditional statutory contrast, not the tariffs' total causal incidence. Across two loss anchors and three duration-equivalent scales, wage-cut cushioning exceeds displacement cushioning by 9.8–12.8 percentage points. At the unit twelve-month scale the difference is 10.0 points and the wage-cut Exchequer cost is £107 million higher. Proportional cuts receive tax relief near marginal rates, while whole-job loss receives relief nearer average rates and relies more on UC. Part of the wage-cut difference is lower pension saving under the HBAI cash-income measure, not current tax-benefit insurance.

Balanced integration makes the common aggregate loss comparable assignment by assignment and closely reproduces the Bernoulli mean at unit scale. It does not overcome sparse information: the unit displacement loss has only 6.2 effective record contributions, and support is weaker at the smallest OBR-duration cells. Assignment SDs therefore describe scenario-allocation sensitivity, not population confidence intervals. The sign also survives all 23 leave-one-division-out reruns, with a cushioning difference of 7.6–11.3 points. This establishes sector robustness within the imposed bridge, not causal robustness of that bridge.

Calibrated longitudinal-LFS allocation sensitivities leave the central extensive-margin conclusion broadly unchanged: cushioning spans 36.0–37.4 per cent across cell, earnings-band and regularised-QRF risk shapes, compared with 36.8 per cent under uniform assignment. This robustness is conditional and limited. The imputation models agree weakly person by person, and the available LFS manufacturing target has only 177 donors; stability of an aggregate fiscal ratio is not evidence that tariff-specific worker selection has been identified.

National poverty changes remain near zero because the exposed population is small and often above the poverty line. They are not precise population estimates and do not imply that affected households' losses are small. EPD, inactivity, reallocation and demographic results are secondary or supplementary rather than competing headline estimands.

The main limitation is structural: the model starts at the end of the production-side transmission mechanism. It does not estimate how the tariff changes border prices, export quantities, domestic output, value added, imported-input costs, firm productivity, investment, profits or labour demand. It assigns calibrated exposure directly to the wage bill and then assumes how that loss is divided among wages, employment, inactivity and reallocation. The results therefore answer a conditional fiscal-incidence question—what statutory taxes and transfers do after specified labour-income changes—not what the tariff does to the economy. Assignment is also independent within broad SIC divisions. The new longitudinal LFS exercise shows that plausible cell and forest incidence shapes agree only weakly even after matching aggregate targets; with just 177 manufacturing donors it cannot identify tariff-specific worker selection. The inactivity scenario assumes WCA passage for selected older workers; New Style JSA and within-year timing are absent; household labour-supply responses are suppressed; and indirect taxes, exchange rates, market switching, retaliation and general equilibrium are outside the central model. The supply-chain calculation is an accounting sensitivity, not a lower bound, and cannot be mechanically combined with local-employment multipliers. Constituency results in the online supplement are illustrative because local industry is not observed at the required level.

The companion UK AI study suggests useful extensions to the analysis architecture: separate the aggregate shock size from its incidence pattern as factorial scenario axes; add mixed wage–employment adjustment rather than only polar margins; benchmark exposure against external employment data; distinguish literature-calibrated from measured-incidence scenarios; evaluate policy counterfactuals on common draws; and generate paper values from a declared build manifest. Those extensions improve transparency and robustness, but they do not replace the missing trade-to-production first stage. The priority for a substantive extension is therefore a product- or firm-level model linking tariffs to prices, quantities, output, value added and productivity before any household microsimulation is run.

The defensible conclusion is therefore modest. For a given direct earnings loss, the adjustment margin materially changes the mix of tax relief, benefit support, poverty and fiscal cost. The polar margins bracket a range of adjustment assumptions, and the 15/85 mixed result is a literature-disciplined sensitivity rather than an estimated decomposition. The close aggregate match between the reference and observed export falls does not validate the bridge because sector agreement is weak. Quantifying the realised tariff effect requires causal trade and linked employer–worker evidence beyond this microsimulation.

Data and code availability

The analysis code, configuration, tests, input manifest, and generated result artifacts are available in the accompanying GitHub repository. The manifest records source URLs, retrieval dates, vintages, exclusions, and SHA–256 hashes. The Family Resources Survey and certain ONS supply-chain workbooks are licensed or access-controlled and are therefore not redistributed. A fully fresh reproduction requires those files (or the documented cached inputs), the Python environment specified by `uv.lock`, and the commands in the repository's `Makefile`. Continuous integration validates the public code path and manifest; it cannot validate inputs that cannot legally be distributed.

Declarations

The author reports no conflicts of interest. No external funding was received for this study. The project uses secondary, de-identified survey and administrative-source inputs; no new human-participant data were collected. The author is responsible for the analysis, interpretation, and any remaining errors.

References

Office for National Statistics (2025). *Labour Force Survey Five-Quarter Longitudinal Dataset, July 2024–September 2025*. UK Data Service, Study Number 9490. doi:10.5255/UKDA-SN-9490-1.

HM Revenue and Customs (2026). *UK overseas trade in goods statistics: detailed commodity by country data*. HMRC Online Trade Statistics API, monthly observations through May 2026. https://www.uktradeinfo.com/.

ABPI (2025). UK–US agreement on pharmaceutical tariffs and pricing. Association of the British Pharmaceutical Industry, London, December 2025.

Acemoglu, D., Akcigit, U. and Kerr, W. (2016). Networks and the macroeconomy: An empirical exploration. *NBER Macroeconomics Annual*, 30(1), 273–335.

Acosta, M. and Cox, L. (2024). The regressive nature of the US tariff code: Origins and implications. Working paper, February 2024 (conditionally accepted, *The Quarterly Journal of Economics*).

Adão, R., Arkolakis, C. and Esposito, F. (2023). General equilibrium effects in space: Theory and measurement. NBER Working Paper No. 25544, National Bureau of Economic Research, Cambridge, MA.

Amiti, M., Redding, S. J. and Weinstein, D. E. (2019). The impact of the 2018 tariffs on prices and welfare. *Journal of Economic Perspectives*, 33(4), 187–210.

Aragón, F. M., Rud, J. P. and Toews, G. (2018). Resource shocks, employment, and gender: Evidence from the collapse of the UK coal industry. *Labour Economics*, 52, 54–67.

Artuc, E., Porto, G. and Rijkers, B. (2021). Household impacts of tariffs: Data and results from agricultural trade protection. *The World Bank Economic Review*, 35(3), 563–585.

Auerbach, A. J. and Feenberg, D. (2000). The significance of federal taxes as automatic stabilizers. *Journal of Economic Perspectives*, 14(3), 37–56.

Atkin, D., Bernadac, V., Donaldson, D., Garg, T. and Huneeus, F. (2025). The incidence of distortions. Working paper, December 2025 draft.

Auray, S., Devereux, M. B. and Eyquem, A. (2020). Trade wars, currency wars. NBER Working Paper No. 27460, National Bureau of Economic Research, Cambridge, MA.

Autor, D. H., Dorn, D. and Hanson, G. H. (2013). The China syndrome: Local labor market effects of import competition in the United States. *American Economic Review*, 103(6), 2121–2168.

Autor, D. H., Dorn, D., Hanson, G. H. and Song, J. (2014). Trade adjustment: Worker-level evidence. *The Quarterly Journal of Economics*, 129(4), 1799–1860.

Autor, D. H., Dorn, D. and Hanson, G. H. (2021). On the persistence of the China shock. *Brookings Papers on Economic Activity*, Fall 2021, 381–447.

Bank of England (2025). *Monetary Policy Report*, May, August and November 2025 issues. Bank of England, London.

Barrot, J.-N. and Sauvagnat, J. (2016). Input specificity and the propagation of idiosyncratic shocks in production networks. *The Quarterly Journal of Economics*, 131(3), 1543–1592.

Beatty, C. and Fothergill, S. (2017). The impact on welfare and public finances of job loss in industrial Britain. *Regional Studies, Regional Science*, 4(1), 161–180.

Beatty, C., Fothergill, S. and Powell, R. (2007). Twenty years on: Has the economy of the UK coalfields recovered? *Environment and Planning A*, 39(7), 1654–1675.

Beatty, C. and Fothergill, S. (2023). The persistence of hidden unemployment among incapacity claimants in large parts of Britain. *Local Economy*, doi:10.1177/02690942231184815.

Blanchard, E. J., Bown, C. P. and Chor, D. (2019). Did Trump's trade war impact the 2018 election? NBER Working Paper No. 26434, National Bureau of Economic Research, Cambridge, MA.

Bloom, N., Bunn, P., Chen, S., Mizen, P., Smietanka, P. and Thwaites, G. (2019). The impact of Brexit on UK firms. NBER Working Paper No. 26218, National Bureau of Economic Research, Cambridge, MA.

Borusyak, K. and Jaravel, X. (2021). The distributional effects of trade: Theory and evidence from the United States. NBER Working Paper No. 28957, National Bureau of Economic Research, Cambridge, MA.

Bourguignon, F., Bussolo, M. and Pereira da Silva, L. A. (eds.) (2008). *The Impact of Macroeconomic Policies on Poverty and Income Distribution: Macro-Micro Evaluation Techniques and Tools*. Palgrave Macmillan and World Bank, Washington, DC.

Brewer, M. and Tasseva, I. V. (2021). Did the UK policy response to Covid-19 protect household incomes? *The Journal of Economic Inequality*, 19(3), 433–458.

Breinlich, H., Leromain, E., Novy, D. and Sampson, T. (2022). The Brexit vote, inflation and UK living standards. *International Economic Review*, 63(1), 63–93.

The Budget Lab (2025). Where we stand: The fiscal, economic, and distributional effects of all U.S. tariffs enacted in 2025. The Budget Lab at Yale, New Haven, CT.

Caliendo, L., Dvorkin, M. and Parro, F. (2019). Trade and labor market dynamics: General equilibrium analysis of the China trade shock. *Econometrica*, 87(3), 741–835.

Card, D., Cardoso, A. R., Heining, J. and Kline, P. (2018). Firms and labor market inequality: Evidence and some theory. *Journal of Labor Economics*, 36(S1), S13–S70.

Cavallo, A., Gopinath, G., Neiman, B. and Tang, J. (2021). Tariff pass-through at the border and at the store: Evidence from US trade policy. *American Economic Review: Insights*, 3(1), 19–34.

Cavallo, A., Llamas, P. and Vazquez, F. (2025). Tracking the short-run price impact of U.S. tariffs. NBER Working Paper No. 34496, National Bureau of Economic Research, Cambridge, MA.

Centre for Economic Performance (2025). UK firms and the 2025 US tariffs: Evidence from the Decision Maker Panel. VoxEU column, Centre for Economic Policy Research, and CEP "in brief" 11639, Centre for Economic Performance, LSE, London. https://cepr.org/voxeu/columns/impact-2025-us-tariff-announcements-uk-firms.

Centre for Cities (2025). The uneven storm: How US tariffs hit UK cities differently. Centre for Cities, London.

City-REDI (2025). The regional impact of US automotive tariffs on the UK economy. City-REDI, University of Birmingham.

Corsetti, G., Lloyd, S. and Ostry, D. (2025). Trading blows: The exchange-rate response to tariffs and retaliations. Staff Working Paper No. 1,139, Bank of England, London.

Dauth, W., Findeisen, S. and Suedekum, J. (2021). Adjusting to globalization in Germany. *Journal of Labor Economics*, 39(1), 263–302.

De Lyon, J. and Pessoa, J. P. (2021). Worker and firm responses to trade shocks: The UK–China case. *European Economic Review*, 133, 103678.

Department for International Trade (2021). *Trade Modelling Review Expert Panel: Report and Recommendations*. Department for International Trade, London.

den Besten, T. and Känzig, D. R. (2025). The macroeconomic effects of tariffs: Evidence from U.S. historical data. NBER Working Paper No. 34852, National Bureau of Economic Research, Cambridge, MA.

Dhingra, S. (2025). Tariffed with the same brush. Speech, Bank of England, October 2025.

Dhingra, S., Ottaviano, G., Sampson, T. and Van Reenen, J. (2017). The costs and benefits of leaving the EU: Trade effects. *Economic Policy*, 32(92), 651–705.

Dix-Carneiro, R. (2014). Trade liberalization and labor market dynamics. *Econometrica*, 82(3), 825–885.

Dix-Carneiro, R. and Kovak, B. K. (2017). Trade liberalization and regional dynamics. *American Economic Review*, 107(10), 2908–2946.

Dolls, M., Fuest, C. and Peichl, A. (2012). Automatic stabilizers and economic crisis: US vs. Europe. *Journal of Public Economics*, 96(3–4), 279–294.

Dorn, D. and Levell, P. (2024). Labour market impacts of the China shock: Why the tide of globalisation did not lift all boats. *Labour Economics*, 91, 102629.

Department for Work and Pensions (2024). Income-related benefits: Estimates of take-up. Official statistics series, DWP, London.

Fajgelbaum, P. D. and Khandelwal, A. K. (2016). Measuring the unequal gains from trade. *The Quarterly Journal of Economics*, 131(3), 1113–1180.

Fajgelbaum, P. D. and Khandelwal, A. K. (2022). The economic impacts of the US–China trade war. *Annual Review of Economics*, 14, 205–228.

Fajgelbaum, P. D. and Khandelwal, A. K. (2026). Tariffs in 2025: Short-run impacts on the U.S. economy. *Brookings Papers on Economic Activity*, Spring 2026 conference draft.

Fajgelbaum, P. D., Goldberg, P. K., Kennedy, P. J. and Khandelwal, A. K. (2020). The return to protectionism. *The Quarterly Journal of Economics*, 135(1), 1–55.

Fetzer, T. (2019). Did austerity cause Brexit? *American Economic Review*, 109(11), 3849–3886.

Fetzer, T. and Wang, S. (2024). Measuring the regional economic cost of Brexit: Evidence up to 2019. CAGE Working Paper No. 1280, University of Warwick.

Flaaen, A. and Pierce, J. R. (2019). Disentangling the effects of the 2018–2019 tariffs on a globally connected U.S. manufacturing sector. Finance and Economics Discussion Series 2019-086, Board of Governors of the Federal Reserve System, Washington, DC.

Galle, S., Rodríguez-Clare, A. and Yi, M. (2023). Slicing the pie: Quantifying the aggregate and distributional effects of trade. *The Review of Economic Studies*, 90(1), 331–375.

Garin, A. and Silvério, F. (2024). How responsive are wages to firm-specific changes in labour demand? Evidence from idiosyncratic export demand shocks. *The Review of Economic Studies*, 91(3), 1671–1710.

Gathmann, C., Helm, I. and Schönberg, U. (2020). Spillover effects of mass layoffs. *Journal of the European Economic Association*, 18(1), 427–468.

Hummels, D., Jørgensen, R., Munch, J. and Xiang, C. (2014). The wage effects of offshoring: Evidence from Danish matched worker–firm data. *American Economic Review*, 104(6), 1597–1629.

Hyman, B., Kovak, B. K. and Leive, A. (2024). Wage insurance for displaced workers. NBER Working Paper No. 32464, National Bureau of Economic Research, Cambridge, MA.

Institute for Fiscal Studies (2024). Health-related benefit claims post-pandemic: UK trends and global context. IFS Report, Institute for Fiscal Studies, London, September 2024.

International Monetary Fund (2025). *World Economic Outlook: Global Economy in Flux, Prospects Remain Dim.* IMF, Washington, DC, October 2025.

Ignatenko, A., Lashkaripour, A., Macedoni, L. and Simonovska, I. (2025). Making America great again? The economic impacts of Liberation Day tariffs. *Journal of International Economics*, 157, 104138.

Irastorza-Fadrique, A., Levell, P. and Parey, M. (2025). Household responses to trade shocks. *Journal of International Economics*, forthcoming (IFS Working Paper 24/52).

Jeanne, O. and Son, J. (2024). To what extent are tariffs offset by exchange rates? *Journal of International Money and Finance*, 142.

Kalemli-Özcan, Ş., Soylu, C. and Yildirim, M. A. (2026). Global trade, tariff uncertainty and the U.S. dollar. NBER Working Paper No. 34728, National Bureau of Economic Research, Cambridge, MA.

Kim, R. and Vogel, J. (2021). Trade shocks and labor market adjustment. *American Economic Review: Insights*, 3(1), 115–130.

Kline, P., Petkova, N., Williams, H. and Zidar, O. (2019). Who profits from patents? Rent-sharing at innovative firms. *The Quarterly Journal of Economics*, 134(3), 1343–1404.

Levell, P., O'Connell, M. and Smith, K. (2017). The exposure of households' food spending to tariff changes and exchange rate movements. IFS Briefing Note BN213, Institute for Fiscal Studies, London.

Luu, N., Woloszko, N., Causa, O., Arriola, C., van Tongeren, F. and Johansson, Å. (2020). Mapping trade to household budget survey: A conversion framework for assessing the distributional impact of trade policies. OECD Trade Policy Papers No. 244, OECD Publishing, Paris.

McKay, A. and Reis, R. (2016). The role of automatic stabilizers in the U.S. business cycle. *Econometrica*, 84(1), 141–194.

Michelena, G., Ernst, C. and Bertin, P. (2025). Tariffs and labor markets: The employment impact of the recent trade conflict. A multiregional input–output analysis. International Labour Organization / arXiv:2512.11578, December 2025.

Moretti, E. (2010). Local multipliers. *American Economic Review: Papers & Proceedings*, 100(2), 373–377.

Navicke, J., Rastrigina, O. and Sutherland, H. (2013). Nowcasting indicators of poverty risk in the European Union: A microsimulation approach. EUROMOD Working Paper No. EM11/13, ISER, University of Essex.

NIESR (2024). Implications of higher US tariffs for the UK economy. National Institute of Economic and Social Research, London, December 2024.

Office for Budget Responsibility (2025). *Economic and Fiscal Outlook*, March 2025 (tariff scenario box). OBR, London.

Office for National Statistics (2025). UK trade with the United States: 2024. ONS, Newport, 25 April. https://www.ons.gov.uk/economy/nationalaccounts/balanceofpayments/articles/uktradewiththeunitedstates/2024.

Peichl, A. (2016). Linking microsimulation and CGE models. *International Journal of Microsimulation*, 9(1), 167–174.

Resolution Foundation (2025a). Trump tariff turmoil. Resolution Foundation, London, April 2025.

Resolution Foundation (2025b). Saving penalties: Reforming the capital rules in Universal Credit. Resolution Foundation, London, April 2025.

Roberts, J. and Taylor, K. (2022). New evidence on disability benefit claims in Britain: The role of health and the local labour market. *Economica*, 89(353), 131–160.

Society of Motor Manufacturers and Traders (2025). Written submission to the House of Commons Business and Trade Committee inquiry on UK–US trade (reference UST0041). UK Parliament written evidence, https://committees.parliament.uk/writtenevidence/167352/html/.

Rodríguez-Clare, A., Ulate, M. and Vasquez, J. P. (2025). The 2025 trade war: Dynamic impacts across U.S. states and the global economy. NBER Working Paper No. 33792, National Bureau of Economic Research, Cambridge, MA.

Rud, J. P., Simmons, M., Toews, G. and Aragón, F. (2022). Job displacement costs of phasing out coal. IZA Discussion Paper No. 15581.

Sutherland, H. and Figari, F. (2013). EUROMOD: The European Union tax–benefit microsimulation model. *International Journal of Microsimulation*, 6(1), 4–26.

Tax Policy Center (2025). Description of the Tax Policy Center's tariff models. Urban–Brookings Tax Policy Center, Washington, DC.

Traiberman, S. (2019). Occupations and import competition: Evidence from Denmark. *American Economic Review*, 109(12), 4260–4301.

Waugh, M. E. (2019). The consumption response to trade shocks: Evidence from the US–China trade war. NBER Working Paper No. 26353, National Bureau of Economic Research, Cambridge, MA.

World Trade Organization (2025). *Global Trade Outlook and Statistics: Update October 2025.* WTO, Geneva.